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RESEARCH ARTICLE

Enhancing Sports Team Management Through Machine Learning

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ABSTRACT Sports like basketball and baseball have seen significant improvements through the effective use of sports analytics. However, research on machine learning in football has primarily focused on prediction. There is a need to explore whether machine learning can contribute to generating more in-depth and accurate insights into football analysis. This essay presents a descriptive analysis of player performance in football matches. Typically, player performance in football is evaluated by consulting a panel of experts, though the criteria they use for evaluation remain undisclosed. By applying machine learning algorithms and pruning techniques, several classifications were developed to identify the most critical functional attributes of players. The functional metrics that delivered the best results in rating predictions had an average error of 0.17. The most significant performance data for each position on the field were closely aligned with the factors experts consider when grading players. The 34 performance characteristics identified achieved a prediction accuracy of 63.4% for match outcomes.

INDEX TERMS Machine learning, sports teams, sports analysis.

I. INTRODUCTION

Sports analysis is the use of scientific methods to the study and modeling of professional athletes' performances in sports [1], [2], [3], [4]. The concepts and methods of game theory [8], [9], biomechanics [10], [11], [12], [13], [14], kinesiology [12], [15], [16], data mining [5], [6], [7], and statistics were often used in this subject. Basketball and baseball have benefited from the use of sports analytics. Despite its widespread global popularity, football has relied heavily on the application of sports analysis. The fundamental cause is the widespread lack of knowledge regarding football. Manchester City Football Club and data company OPTA have teamed to make an enormous dataset for the 2011–2012 Premier League season accessible to the public. This dataset includes comprehensive information for each player in every Premier League match, which will aid in the field of football analytics.

One of the finest sports for collaboration is football. Even while a team's success is often measured by its victories and losses, these results are often the consequence of intricate and intriguing interplay between the players' performances on the two sides as well as an unstable situation. One might even argue that football is a beautiful game because of these exchanges and the inherent ambiguity of the sport [17], [18]. Selecting the right collection of variables to assess player performance is challenging because of the intricate nature of the interaction. At the moment, teams and players are awarded points by common football experts instead of having their performances assessed and compared. Sports media and websites carry these rankings. Football fans love player ratings so much that they gave rise to a game called fantasy football, where players construct an imagined squad of real football players and award points according to how well those players really perform. On the other hand, players are ranked among football specialists according to their knowledge and expertise in the game. Notably, the experts do not examine the specific standards that are applied to the "black box" utilized to determine this rating. It is possible to argue that professionals in these fields use a concealed knowledge that is hard to pin down. One of the aims of this paper is to make this concealed information clear. This study makes use of both the soccer statistics website WhoScored.com and the data gathered by OPTA. However, thus far, research has been

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limited to signal analysis from football matches due to the inherent structure of the available data, which comprises a combination of tactical and technical performance indicators. Using player performance data and machine learning, the primary objectives of this study are: 1. Determining which player performance traits are most crucial in influencing the experts' judgments. In this manner, gaining access to the expertise of professionals in player evaluation. 2. Determining whether players' performances from two rival teams have an impact on the games' outcomes and how much of an impact their performances have on the games' outcomes.

Rahman uses DNNs and ANNs to predict football match outcomes, achieving 63.3% accuracy for FIFA World Cup 2018 matches. It highlights the potential of deep learning in sports analytics but emphasizes the need for more accurate and comprehensive datasets to improve predictions. The study's reliance on data quality and limited generalizability are key limitations [19]. The paper by Davis et al. outlines the challenges of using machine learning to create effective indicators and models from the vast and complex data available in sports, aiming to bridge the gap between traditional sports expertise and modern data analytics. It highlights key methodological issues, such as dependencies in data partitioning and the importance of contextual factors, as well as the distinction between evaluating the indicators and the models behind them. The strengths of the paper include providing a structured framework with practical examples to guide decision-making in professional sports clubs. However, the paper's main limitation is that while it identifies critical challenges and offers insights, it may not delve deeply into specific, actionable solutions for all types of sports data scenarios, potentially leaving gaps in implementation for practitioners [20]. Han and Lee use LDA topic modeling on 1,863 papers from 1990 to 2022 to identify trends in soccer psychology research. It highlights the prominence of topics like ''Coaching Essentials in Football'' and the decline of research on ''Psychological Skills for Performance Development,'' while noting that ''Motivation, cognition, and emotion'' is an underexplored area. The paper effectively shows trends and research gaps but could improve by offering more detailed recommendations for expanding research and practical applications for coaching and player performance [21]. Papageorgiou et al. benchmarked 14 machine learning models for basketball performance forecasting using 18 advanced statistics and KPIs on 90 high-performance players. They found that tree-based models, especially Extra Trees, performed best, achieving a WAPE of 34.14% and a 3.6% MAPE improvement on unseen data. The study effectively highlights the potential of ML in sports analytics but could benefit from more explanation on why tree-based models outperformed others and a discussion on the potential limitations of using only high-performance players, which may impact generalizability [22].

This study introduces novel methodologies to enhance football performance analysis by leveraging machine learning and advanced feature selection techniques. Unlike prior research focused primarily on prediction, this work uncovers hidden criteria used by expert evaluators and identifies critical performance metrics for each player position with high accuracy. Furthermore, it proposes an innovative aggregation method for team-level scores, utilizing a weighted averaging strategy that prioritizes recent performance to improve predictive accuracy. By integrating machine learning approaches with player statistical data, the study not only evaluates individual player performance but also explores how these performances impact match outcomes. This approach offers a fresh perspective on understanding expert rankings while demonstrating the potential of data-driven models to provide deeper insights into football analytics. As statistical and computational methods continue to transform sports performance analysis, this work contributes valuable tools for coaches, athletes, and management to enhance performance comparison, match outcome prediction, and visualization.

II. FINDING CRITERIA FOR SCORES OF FOOTBALL EXPERTS

Two eleven-man teams compete against one another on a rectangular field with a goal on each side while carrying a spherical object in soccer. The objective of each team is to deliver the soccer ball into the goal of the opposing team, which is also referred to as a ''goal.'' Eleven players comprise each team, with one serving as the goalkeeper and the remaining ten occupying positions as strikers, defenders, or midfielders. Goalkeepers are the sole individuals authorized to make physical contact with the ball using their arms or hands within their penalty area. The remaining 10 players on the squad are limited to shooting or passing the ball with their feet, heads, or hips. The squad with the most goals scored was deemed the victor. Depending on the format of the tournament, if both sides have the same number of goals at the conclusion of the game, the result will either be a draw or extra time with penalty kicks.

A. DATA PREPARATION

10,369 samples and 210 attributes make up the player performance dataset for the English Premier League that OPTA released. A player's performance in a match is represented by each sample. Out of the 210 player characteristics included in this dataset, 198 were associated with performance statistics, while the other attributes comprised player profiles specific to that particular match. For every match, expert ratings are necessary for every player's performance. WhoScored.com is used to obtain the expert rating for every player in every game. Please take note that the player ratings on this website are also determined using OPTA information. If this weren't the case, it would likely be harder to uncover the expertise of specialists. Lohnstein's distance between the players' names was employed to apply the relationship in the data since there were discrepancies in certain players' names between the WhoScored.com and OPTA portal data. Consequently, an expert rating dataset was produced. The expert rating

dataset and the OPTA dataset were then combined. As of right now, every record in our dataset includes an expert rating from WhoScored and a player's performance metrics from OPTA, which show how the player fared in a particular Premier League match. A football player is able to perform four different positions: goalie, midfielder, defender, and striker. Given the differences between these four positions, different standards need to apply when determining which players should get expert ratings for certain positions. As a result, we separate our data set into four categories: goalkeepers, midfielders, defenders, and strikers.

B. METHOD

For four soccer posts, these are the four pieces of data. The goal class or expert rating and the participants' performance attributes are used to classify each data set. The Weka software package's algorithms have been used to categorization tasks. Vaka comprises machine learning techniques designed for data mining applications. Many classification methods are used to its data set, and many tests are carried out with the assistance of various factors. The bias resulting from a particular sample of the training data as well as the differences between the training and test sets are minimized by the use of cross-classified validation. The employed classification algorithms consist of discrete regression using J48, decision table, multilevel perceptron, simple linear regression, local weighted learning, IBk, KStar, and RBF network, and linear regression using SMOreg, Gaussian processes, least median linear regression, M5P, Tree Bagging REP, incremental regression, and Tree REP. It is permissible to utilize the default parameters of the algorithms, except for the group algorithms. Four data sets are subjected to a series of categorization methods, and the outcomes are verified. Additionally, we use many features from the data set by using three distinct methodologies, and the effectiveness of the categorization is evaluated and contrasted. The first pruning technique, known as global sequential pruning, involves first sorting the features according to the best algorithms' given weights in decreasing order. Next, step 5 progressively selects the features from the end of the list. In the second pruning approach, the number of features is decreased in the next cycle depending on the model that the algorithm developed during its previous iteration. As a result, each algorithm's real-time execution characteristic affects feature pruning. This pruning technique is only useful for algorithms that allow a ranking list of features to be formed by interpreting the resulting model. This indicates that algorithms like the multi-level perceptron cannot use this pruning technique. The third approach, known as threshold pruning, is similar to the first one except that it chooses the features based on a minimal threshold weight. Thus, only characteristics whose weight is less than a certain threshold are chosen by this procedure. The standard deviation and average weights given to the model's features that were discovered during the algorithm's previous iteration are used to calculate this threshold value.

C. RESULT

SMOreg and the linear regression technique perform best across four data sets. Table 1 displays the findings pertaining to the data of the attackers. The correlation coefficient is denoted by CC in this table, along with the mean absolute error (MAE), mean square error (RMSE), relative absolute error (RelAE), and the times for model construction and training (BTime and TTime). We begin with 195 characteristics and use several techniques to reduce them.

TABLE 1. The results of the algorithms with the attackers' data set (195 features).

Algorithm	correlation coefficient	MAE	RMSE	RelAE	BTime (s)	TTime (s)
Linear regression	0.9646	0.1679	0.2379	24.1348	3.9160	31.3670
SMOreg	0.9646	0.1659	0.2386	23.8477	72.0060	520.9650
Gaussian processes	0.9646	0.1682	0.2370	24.1756	25.3690	280.7680
Least median linear regression	0.9639	0.1724	0.2489	24.7807	11.3030	101.0300
M5P	0.9582	0.1806	0.2582	25.9657	4.5400	37.7600
Bagging	0.9145	0.2589	0.3659	37.2189	3.1050	24.8870
Incremental regression	0.8998	0.2965	0.4000	42.6187	0.5250	4.2680
Tree REP	0.8921	0.2934	0.4080	42.1705	0.3190	2.4310
Discrete regression (J48)	0.8697	0.3235	0.4548	46.5085	1.1880	13.0030
Multilevel perceptron	0.7218	0.4975	0.7638	71.5081	461.7650	4167.5960
Simple linear regression	0.7674	0.4295	0.5784	61.7347	0.0310	0.1300
Decision table	0.8489	0.3381	0.4770	48.6047	2.9630	25.1450
Local weighted learning (LWL)	0.7459	0.4365	0.6032	62.7474	0.0010	99.0370
IBk	0.6912	0.4698	0.6936	67.5311	0.0000	3.9460
KStar	0.6422	0.4844	0.7368	69.6353	0.0000	129.9960
RBF network	0.3394	0.6209	0.8485	89.2492	0.4740	5.4930

1) PRUNE GLOBAL ORDER

Examining the model that the linear regression method generated will provide a weighted list of attributes for every player data set. The linear regression approach is used to organize the attributes in this list in decreasing order of absolute weight. This list is meant to be pruned. For example, Table 2 shows the top 20 features from the attacker's post out of the 116 criteria on this list for attackers. In a similar vein, this collection of 94 traits also includes 20 traits exclusive to defenders, which are shown in Table 3. This list includes

62 traits specifically for goalkeepers; Table 5 displays the top 20 attributes for the goalie position. This list includes 126 traits specific to midfielders; Table 4 displays the top 20 traits for this position.

TABLE 2. The 20 important features out of 195 features of the attacking post in linear regression.

Red cards	Error leading to goal position
Mistakes lead to goals	Earning a penalty foul
Last player tackle	Successful air shipments
assist goal	Pulling the ball over the goal line
Missed penalty	Goals during the game
Flowers from the August launch	Goal from a corner kick
A penalty foul has been committed	Key out throw
Successful corners	Goal outside the penalty area
Free kick goal	Fouls committed other than penalties
Spontaneous flowers	Successful ground submissions

TABLE 3. The 20 important features out of 195 features of the defender post in linear regression.

Red cards	Goals during the game
Mistakes lead to goals	Error leading to goal position
Spontaneous flowers	penalty goal
A penalty foul has been committed	Goal from penalty
assist goal	Unsuccessful corners from the right
throw the ball	Goal outside the penalty area
Direct free kick goal	Free kick goal
Pulling the ball over the goal line	Flower
Goal with the left foot	Penalty foul taken
Unsuccessful corners from the left	Last player tackle

TABLE 4. The 20 important features out of 195 features of the midfielder post in linear regression.

Goal from Pratap Out	Goal with the right foot
Goal from a free kick	Goal outside the penalty area
Goals during the game	A goal inside the penalty area
Goal from the corner	Flowers with the left foot
Red card	Whole flowers
Error leading to goal	Direct free kick goal
Penalty foul	Indirect free kick goal
Spontaneous flowers	pass
Goal with a header	Penalty taken
Other flowers	Pulling the ball over the goal line

TABLE 5. The 20 important features out of 195 features of the goalkeeper post in linear regression.

Red cards	Errors committed except the Indian error
Spontaneous flowers	Losing possession of the ball
Blocking the penalty kick	Total number of failed passes
Error leading to goal	A penalty foul has been committed
India made a mistake	Last player tackle
Fouled in the six-yard box	Goals scored outside the penalty area
Penalty foul taken	Blocking shots inside the penalty area
The first flower	Total errors made
The position of the goal in the frame and during the game	to be replaced
Error leading to goal position	pass interception

Using this sorted list of characteristics, attributes are then trimmed for each instance in the player dataset. A 116-dimensional data set with attributes deemed relevant in linear regression is generated for the attackers. Similar to that, a new dataset is formed at the attacker's post and it keeps removing 5 characteristics from the bottom of the sorted list until it only has the top 5 features. The trimmed dataset is generated by repeating the same procedure for goalkeepers, midfielders, and defenders. The machine learning methods are then applied to each of these data sets. All results are meticulously recorded, encompassing the duration devoted to model development and testing, in addition to the absolute mean error, correlation coefficient, mean square error, and relative absolute error. The trends of mean absolute error versus the number of assailant characteristics for all methods are depicted in Figure 1. Individual representations of the average absolute error trend for function-based, rule-based, tree-based, and meta-algorithm groups on the attackers' data are provided in figures 2 (functions) and 3 (tree), respectively. Illustrated are Figures 4 (meta) and 5 (rule and indolent). Additionally, time spent developing the model, training it using various techniques, and adjusting its amount of features are all tracked. Figure 6 displays the model construction time graph for attackers, and Figure 7 displays the model training time.

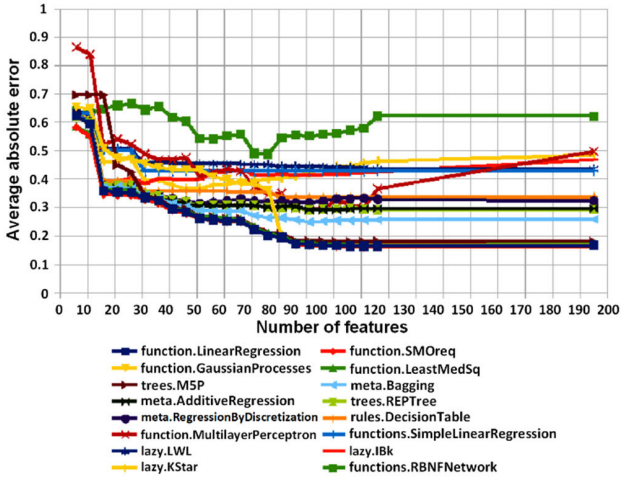


FIGURE 1. Plot of average absolute error changes with increasing number of features.

The SMOreg algorithm yielded the best classification performance in the attackers' data, with an average absolute error of 0.1619 in 111 features. Other algorithms that performed well include linear regression, Gaussian processes, and least median linear regression, which produced an average absolute error of roughly 0.165 in 111 features with little variation in rank. Multi-level perceptron and basic linear regression methods are the exceptions and perform poorly; function-based algorithms (Figure 2) perform the best. It is acceptable to argue that the experts' rating is merely ''number of goals'' as the number of goals scored. The reason for this poor performance is because the linear regression algorithm deems the model too simplistic and only utilizes characteristics to create the model. A player is independent.

The average absolute error of the multi-level perceptron method drops as the number of features is reduced to 86. The multi-level perceptron algorithm with 86 features achieves

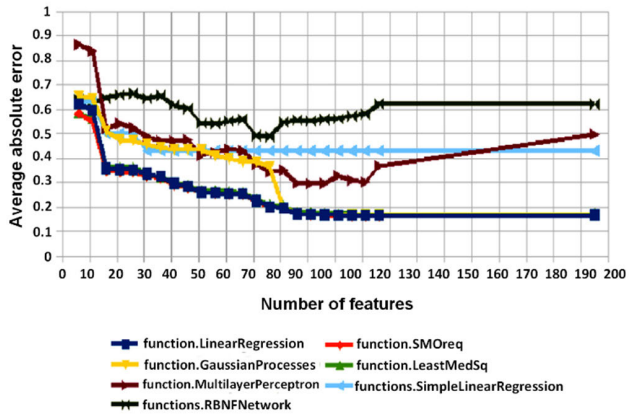


FIGURE 2. Graph of average absolute error changes with increasing number of features in function-oriented algorithms.

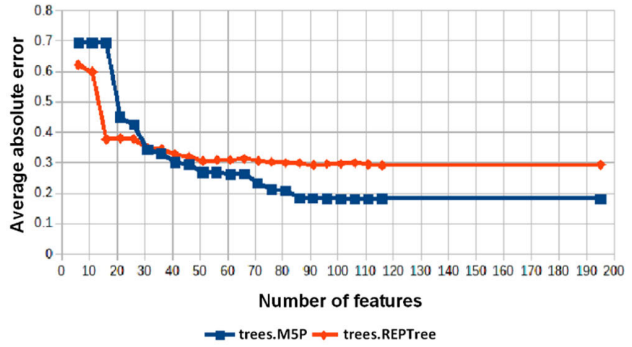


FIGURE 3. Average absolute error changes with increasing number of features in tree algorithms.

the best result, with an average absolute error of 0.29. The inaccuracy grows when the number of characteristics is reduced more. There are about 86 aspects of decreasing error in the techniques shown in Figure 2. Comparing the SMOreg method with 86 characteristics to its greatest performance (with 111 features and an error of 0.1619), the average absolute error is 0.1706, which is a very excellent result. Additionally, Figure 5 shows that when compared to other algorithms, lazy (sample-based) methods have the largest error rate. Examples-based algorithms include the weighted learning algorithm, KStar, which employs a similarity function, IBk algorithm, and closest neighbor classification. It is not always the case that two players with high expert scores have the same performance standards; one player may excel in some areas while the other player has other strong performance traits. A higher score may be achieved using a variety of characteristics. As such, sample-based algorithms are not very effective. The least median linear regression does not vary with the amount of features, IBk has a little change, and the average absolute error of KStar with 46 features is lowered to 0.37 when the average absolute error of these three sample-based methods is analyzed. The distance function that KStar employs and the fact that it is entropy-oriented as opposed to other common algorithms may be the cause of its unusual behavior. The RBF network, which generates basis functions using k-means, performs poorly. Rule-based algorithms, like the decision table algorithm, can provide subpar outcomes (Figure 2). Tree methods (Figure 3) indicate that pruning will not enhance the classification performance of either the REP or M5P tree algorithms. With over 80 characteristics, M5P outperforms the REP tree in terms of performance. While the REP tree is a quick decision tree learner employing information ain or variance, the M5P method creates model trees using a linear regression function in the tree's leaves. Their pruning technique, M5P, is another significant distinction; it prunes by substituting linear regression functions for subtrees [23]. The error reduction approach, also known as the backfitting method, is used by the REP tree to trim the features [24]. The meta-algorithms, which combine the

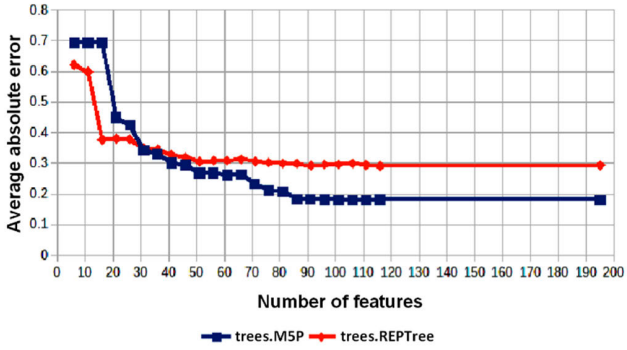


FIGURE 4. Average absolute error changes with increasing number of features in meta-algorithms.

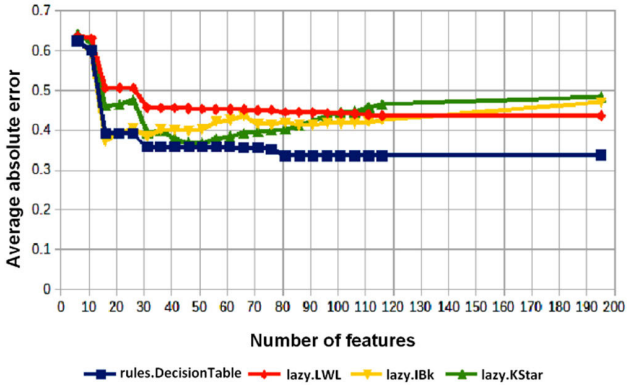


FIGURE 5. Average absolute error changes with increasing number of features in Lazy and Rule based algorithms.

group technique with tree-based algorithms, are shown in Figure 4. In REP tree, incremental regression with decision method, and discrete regression, bagging techniques were used. Tree-based algorithms perform better when paired with ensemble approaches. For instance, the average absolute error of the REP tree is decreased by around 0.04 when bagging is employed in conjunction with it. Pruning features does not enhance performance in meta-algorithms. Similarly, the algorithms behave similarly to attackers when the datasets for goalkeepers, midfielders, and defenders are analyzed. Regarding the collection of useful characteristics for categorization, there are variations throughout different

data sets. The SMOreg method with 94 features yields the smallest average absolute error (0.2673) in the case of defenders, whereas the absolute error in the same algorithm with 69 features is 0.2856. Additionally, the best performance with 69 features is achieved by the multi-level perceptron method, which performs badly in a large number of features. Similarly, a general decrease in mean absolute error is seen at 91 features, the ideal collection of features in midfielders' data, for the midfield position and almost all methods, including the multilayer perceptron technique. This ideal collection has 37 elements for goalies. Table 6 displays the best results across all four data sets, which correspond to four postings. Table 7 displays the results achieved with the ideal amount of characteristics.

TABLE 6. The best results of the global sequential pruning method with the number of non-optimal traits.

Best performance	attackers	defenders	goalkeepers	middle players
Absolute error of mean	0.1619	0.2674	0.2070	0.1708
Algorithm	SMOreg	SMOreg	SMOreg	SMOreg
Number of attributes	111	94	62	126

TABLE 7. The results of the global sequential pruning method with the optimal list of traits.

Best performance	attackers	defenders	Goalkeepers	middle players
Absolute error of mean	0.1706	0.2856	0.2160	0.1763
Algorithm	SMOreg	SMOreg	SMOreg	SMOreg

The multilayer perceptron approach, when a large number of attributes are considered, exhibits a relatively lengthy model construction time (Figure 6) and model training time (Figure 7), when distinct methodologies are employed to monitor these two-time components. To illustrate, the model construction process requires 16 minutes, while the training phase entails 2.5 hours and 195 attributes from the defender's dataset. In contrast, as the number of features decreases, the time necessary to develop and train the model decreases linearly with a 36-degree slope. In addition, Gaussian processes and SMOreg algorithms require considerable time for construction and learning. The SMOreg approach shortens the time required to create and train the model by pruning the features; this is not the case with Gaussian processes, which result in a constant time. When it comes to developing and training models, sample-based algorithms are among of the quickest. Regarding modeling time and training time, a similar pattern is seen in all four data sets.

2) CYCLIC LOCAL PRUNING

SMOreg, Gaussian processes, least median linear regression, M5P, discrete regression and J48, decision tables, perceptrons, incremental regression with decision method, and REP trees are among the linear regression techniques used in each dataset containing 195 characteristics. The

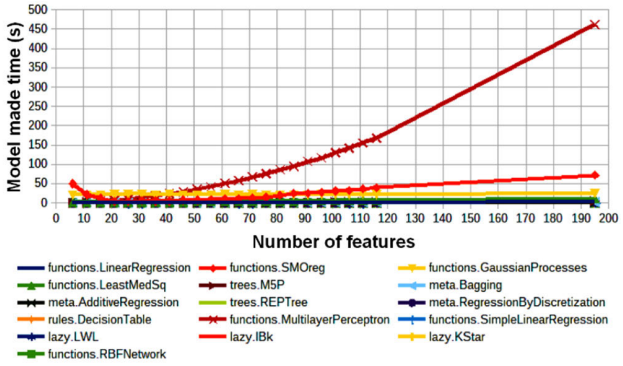


FIGURE 6. Model making time changes with increasing number of features.

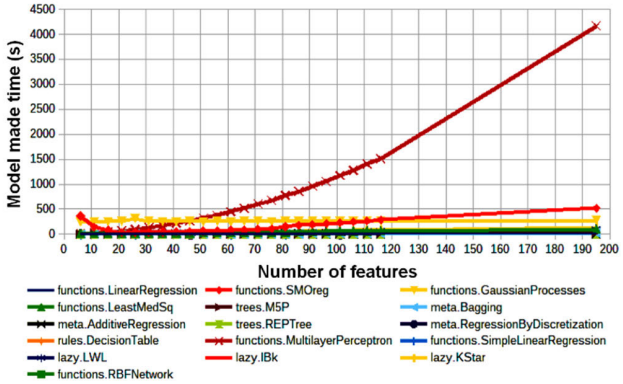


FIGURE 7. Model training time changes with increasing number of features.

techniques used include many layers, weighted local learning, IBk, KStar, RBF, and basic linear regression. Subsequently, algorithms are chosen based on their ability to retrieve a list of characteristics from the categorization models generated. Usually, using interpretable models is the only way to make this job feasible. For local wheel pruning, four techniques are chosen: least median linear regression, M5P, REP tree, and linear regression. The remaining algorithms are tree-based, whereas the first two are functional. The outcomes of Table 2 are obtained by running the algorithms on the data set. This table shows that the M5P tree-based method and function-based algorithms like linear regression and least median linear regression perform well in the data set when it comes to classification. Moreover, models produced by the M5P, REP tree, least median linear regression, and linear regression algorithms may be used to provide an ordered list of attributes. Using this pruning strategy, four algorithms are used to each of the four player data sets, focusing on 195 major characteristics. The first algorithmic cycle extracts the list of features needed to create the model (list A), and the second algorithmic cycle extracts a new data set of existing features. In list A, it is created. Subsequently, the same technique is used to the fresh data set, extracting a list of characteristics that were crucial in building the model and using them to generate the subsequent data set. The attributes that the algorithm employed in the preceding cycle of model building determine the collection of data that is

used to run the algorithm. When the set of features used to build the model in the current cycle does not change from the set used in the previous cycle, pruning is ended at the algorithmic level; this indicates that the pruning process has reached saturation. Table 8 displays the summary of the linear regression algorithm's result for the attacker's data set, whereas Table 9 displays the summary of the least median linear regression algorithm's result. The outcomes of the M5P and REP tree algorithms are shown in Tables 10 and 11, respectively.

TABLE 8. Cyclic local pruning method-linear regression.

number of attributes	Correlation coefficient	MAE	RMSE	RelAE	Model building time (s)	Model test time (s)
195	0.9646	0.1679	0.2379	24.1348	7.4490	60.7150
117	0.9664	0.1644	0.2320	23.6302	0.1610	4.3510

TABLE 9. Cyclic local pruning method - least median linear regression.

number of attributes	Correlation coefficient	MAE	RMSE	RelAE	Model building time (s)	Model test time (s)
195	0.9639	0.1724	0.2489	24.7808	18.6720	159.8560
117	0.9639	0.1724	0.2489	23.6302	16.2680	147.0290

TABLE 10. Cyclic local pruning method - M5P.

number of attributes	Correlation coefficient	MAE	RMSE	RelAE	Model building time (s)	Model test time (s)
195	0.9582	0.1806	0.2582	25.9657	7.7110	49.8840
83	0.9566	0.1826	0.2630	26.2458	2.7460	24.6970
76	0.9577	0.1804	0.2596	25.9324	2.3780	21.7070
64	0.9525	0.1872	0.2747	26.9047	1.7050	16.7360

TABLE 11. Cyclic local pruning method – REP tree.

number of attributes	Correlation coefficient	MAE	RMSE	RelAE	Model building time (s)	Model test time (s)
195	0.8921	0.2934	0.4080	42.1705	1.0360	3.9550
43	0.8911	0.2928	0.4097	42.0959	0.0940	0.7550

It is important to highlight that this pruning technique rapidly approaches saturation. The least median linear regression and REP tree algorithms experience pruning saturation after the second cycle, whereas the linear regression algorithm reaches saturation in the pruning process either after the second or third cycle. Table 12 presents the mean absolute errors made by the algorithms during the saturation cycle, categorized by the number of attributes associated with each of the four positions.

TABLE 12. Cyclic local pruning method - pruning saturation point.

	Attackers	Defenders	Goalkeepers	Middle Players
Linear regression				
Number of attributes	116	94	61	126
Error	0.1644	0.2700	0.2071	0.1722
M5P				
Number of attributes	63	60	34	95
Error	0.1872	0.3086	0.2231	0.1900
Least median linear regression				
Number of attributes	174	173	108	177
Error	0.1724	0.2766	0.2164	0.1764
Tree REP				
Number of attributes	42	42	25	74
Error	0.2928	0.3939	0.3730	0.3232

Table 13 and Table 14 present the outcomes pertaining to threshold pruning when employing a linear regression algorithm and two distinct methods with varying thresholds in the context of attackers. Both threshold values have the same saturation value in the assailants' data, indicating that the optimal set of features is chosen. The initial approach to calculating the threshold (using the mean difference and standard deviation) for defenders and midfielders encounters saturation with a limited number of attributes, resulting in a suboptimal performance compared to the alternative method (which involves multiplying 0.3 by the attribute weight standard deviation). When the mean and standard deviation of the attribute weight at the saturation point are subtracted from the gatekeepers' data, the resulting threshold value exhibits less error and is deemed more optimal in comparison to alternative threshold values. A concise overview of the threshold pruning method's optimal outcomes can be found in Table 15.

In the figure 8, each selected attribute list is contrasted to the other lists. In order to avoid redundant comparisons of similar lists, each list along the horizontal axis is compared exclusively with lists that it has not been compared with previously. An increase in the number of bars along the X axis ensues. The degree of similarity among lists produced by function-based algorithms is clearly the highest.

The final set of attributes for each job that establishes the bachelor's rankings was generated by combining the lists derived from three distinct pruning techniques. A list including every characteristic in Table 16 was made for every football post. These attributes were ranked using the reward and punishment system in the next step. Each attribute was given a weight of n values after its presence in the top n

TABLE 13. Threshold pruning method - linear regression (threshold = average SD).

Number of attributes	Correlation coefficient	MAE	RMSE	RelAE	Model building time (s)	Model test time (s)	Threshold
195	0.9646	0.1679	0.2379	24.1348	7.5490	63.0340	0.0587
39	0.9113	0.2689	0.3715	38.6553	0.0650	0.7590	0.0156
29	0.9103	0.2695	0.3735	38.7333	0.0220	0.2880	0.0381
27	0.9104	0.2693	0.3732	38.7098	0.0180	0.2340	0.0674
28	0.9106	0.2691	0.3728	38.6872	0.0130	0.1690	0.0871

TABLE 14. Threshold pruning method - linear regression (threshold = 0.3 × SD).

Number of attributes	Correlation coefficient	MAE	RMSE	RelAE	Model building time (s)	Model test time (s)	Threshold
195	0.9646	0.1679	0.2379	24.1348	7.3920	58.6910	0.0642
37	0.9113	0.2688	0.3715	38.6448	0.0490	0.6220	0.0778
26	0.9106	0.2691	0.3728	38.6872	0.0190	0.2760	0.0775
25	0.9106	0.2691	0.3728	38.6872	0.0120	0.1680	0.0775

TABLE 15. Optimal results of the threshold pruning method.

Linear regression	Attackers	Defenders	Goalkeepers	Middle players
Number of attributes	25	14	6	13
Error	0.2691	0.4427	0.5718	0.4240

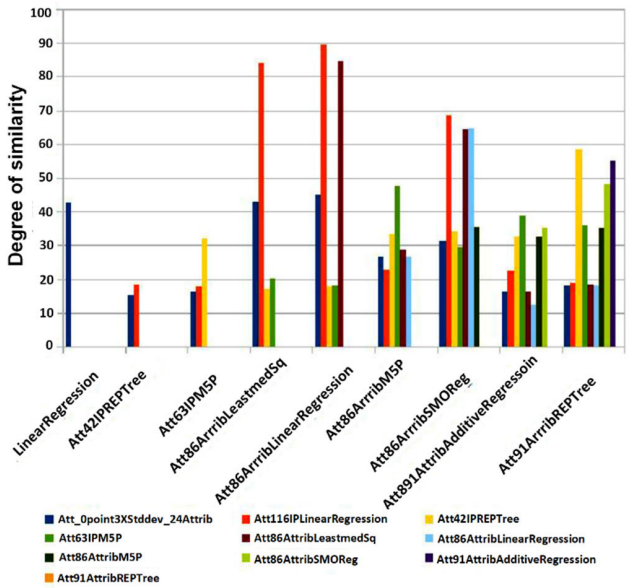


FIGURE 8. The degree of similarity between the optimal lists of attackers.

positions of each list was verified for each attribute's presence in the optimum lists.

TABLE 16. Guide to the set of lists of optimal characteristics of attackers in Figure 8.

Name of the feature list	Name of the algorithm	Number of features	Feature pruning method
Att_0point3XStddev_24Attrib	Linear regression	24	Threshold pruning
Att116IPLinearRegression	Linear regression	116	Cyclic pruning
Att42IPREPTree	Tree REP	42	Cyclic pruning
Att63IPM5P	M5P	63	Cyclic pruning
Att86AttribLeastmedSq	Least median linear regression	86	Orderly nationwide pruning
Att86AttribLinearRegression	Linear regression	86	Orderly nationwide pruning
Att86AttribM5P	M5P	86	Orderly nationwide pruning
Att86AttribSMOReg	SMOreg	86	Orderly nationwide pruning
Att91AttribAdditiveRegression	Linear regression	91	Orderly nationwide pruning
Att91AttribREPTree	Tree REP	91	Orderly nationwide pruning

III. DETERMINING RESULT OF MATCH USING TEAM PERFORMANCE CRITERIA

The goal of this section is to determine which player performance traits have the most impact on the game's result. Next, it is determined if the traits of the players on the two opposing sides have an impact on the outcome of the game. The data set has been generated in the proper formats and has undergone several classifications. The most crucial functional elements are found in the game result detection process by using feature selection.

A. DATA PREPARATION

10,369 samples and 210 attributes make up the player performance dataset for the English Premier League that OPTA released. Among the 210 player traits, 198 were related to performance, while the other ones were used to identify the individual. Both self-reported objectives and match outcomes were absent from the OPTA data collection that was made public. The WhoScored.com website provided the information for each player's individual goals in every game, and this feature was included to the dataset. The players' performance data set was then compressed and altered to provide data for every match that was played since we needed data for every match. After that, the data set was appropriately reduced to only include 380 samples. The performance criteria of each player are combined in that match to create a match data set from their data set. We utilized the total or average aggregation function based on how these attributes affected the team's performance. The performance coefficients of the home and away teams in each match are added together to make a sample in

TABLE 17. Algorithms performance on match data set.

Algorithm	Kappa	F-Measure	ROC	Classification accuracy (%)
Multilevel perceptron	0.5235	0.693	0.836	69.4737
FT	0.5045	0.683	0.801	67.8947
FT with AdaboostM1	0.4939	0.672	0.827	67.6316
SMO	0.4529	0.648	0.754	65.2632
Naive Bayes	0.3384	0.571	0.746	56.5789
Random Forest	0.3094	0.552	0.74	56.8421
Decision Table	0.3196	0.554	0.724	57.6316
FURIA	0.3162	0.551	0.715	56.8421
J48Graft	0.275	0.527	0.637	53.9474
J48	0.2619	0.522	0.631	52.6316
Jrip	0.2174	0.48	0.622	52.1053

TABLE 18. Performance of algorithms with 34 selected features on the match dataset.

Algorithm	Kappa	F-Measure	ROC	Classification accuracy (%)
Multilevel perceptron	0.3694	0.593	0.783	59.4737
FT	0.4098	0.619	0.736	62.1053
SMO	0.4206	0.621	0.74	63.4211
Naive Bayes	0.3154	0.555	0.761	54.7368
Random Forest	0.3064	0.549	0.743	56.3158
Decision Table	0.3385	0.566	0.747	58.6842
FURIA	0.287	0.501	0.501	56.8421
J48Graft	0.2591	0.522	0.635	52.6316
J48	0.2659	0.527	0.644	52.8947
Jrip	0.2399	0.493	0.636	54.2105
REP Tree	0.2708	0.513	0.668	55
LibSVM	0.3667	0.279	0.5	45
Kstar	0.2438	0.512	0.7	51.5789

the newly created dataset, which represents a match. As a result, the dataset's feature count doubled. The data was then examined using dimensionality reduction techniques. The dataset was cleansed of plug-in characteristics that were the same for every player. Subsequently, a few characteristics pertaining to matches were included into the dataset, resulting in 368 features for every game instance. The second phase was creating another data table that had 20 samples for each team, and it included an overview of each team's performance over the course of the whole season. This data set's numbers are compared to those found on the sports website for each team's total goals scored, wins, and losses. Your data's validity and correctness are shown in this manner. For testing purposes, player and match level statistics are currently being utilized.

B. RESULT

Supervised classification aims to forecast the outcome of a game or the target class by utilizing the performance of two teams and the "home team victory" criterion. There are three potential outcomes for the match. All extant features that have a direct impact on the game's outcome, including "draw" and "home team defeat" as well as those that specify scored goals, are eliminated. For instance, the aggregate number of goals scored by the home or away team from corners, free kicks, penalties, clean sheets for either the home or away team, and so forth. In order to prevent the establishment of overt regulations governing the outcome of the match, only goals scored and goals received can be considered decisive factors. Consequently, any attributes associated with these two components are omitted. Subsequently, the feature count of the dataset further diminishes to 336 in number. Subsequently, a number of machine learning algorithms are executed on the competition data set, and the impact of feature selection on the target class, which is the prediction of competition results, is observed.

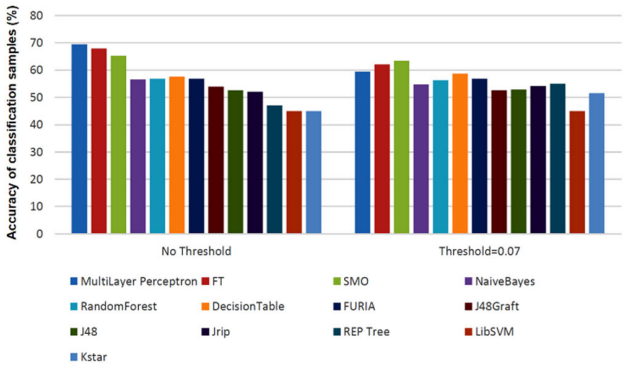


FIGURE 9. Comparison of Algorithms: Accuracy percentage of classified samples against Gain-Ratio threshold.

IV. PLAYER SCORE COMBINATION

This section evaluates the correlation between player rankings and match results and their predictive power for future matches. The process involves finding methods to combine player rankings into team rankings as performance indicators, analyzing the system's effectiveness in predicting current match outcomes, and using it to forecast future match results based solely on past player performance rankings. The experiment has two main goals: First, to develop methods for aggregating player rankings into team rankings that effectively predict match outcomes, using team rankings as input for classification. Two hypotheses are tested—whether expert rankings reflect only player performance (with the assumption the best team wins) or are influenced by match results (with a higher likelihood of being true). The second objective is to evaluate the accuracy of predicting the next match outcome using team rankings generated from the aggregation method, focusing on player performance in past matches. Figure 10 shows a flowchart for predicting the outcome of a match using the performance scores of players in the current match.

The experiment modifies a dataset of 10,369 samples, each representing a player's performance in English Premier League matches. Initially containing 195 performance metrics, including expert rankings, the dataset is expanded with match-level details (e.g., date, teams, venue) and player-specific attributes (e.g., name, position). The final

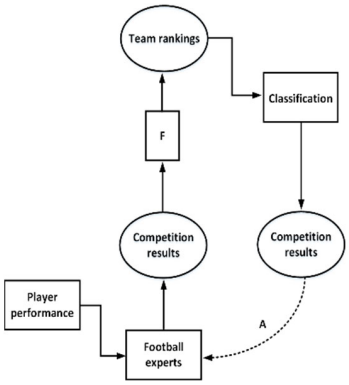


FIGURE 10. Flowchart for predicting the outcome of a match.

dataset comprises 202 features. Classification algorithms such as SMO, LibSVM, Bagging, and AdaBoost were applied to a dataset of 27 aggregated player ranking features to predict match outcomes, achieving high accuracy. Feature importance analysis revealed that average team rankings and position-specific averages were critical. Additional derived features, such as differences in team and position-specific scores, were introduced, leading to improved performance. For example, using differences in average rankings and position-specific scores achieved 88.68% accuracy with SMO, while incorporating a mix of averages and extremes (e.g., maximum and minimum scores by position) improved accuracy to 89.21%. Various feature subsets were tested to identify the optimal predictors.

After several iterations with different subsets of algorithms, we concluded that instead of using the difference between the average rankings of the teams, it is better to use the average score of the home team and the average score of the away team as two separate features. However, for position-specific average rankings, such as the average score for forwards of the home team or the away team, it is better to use the difference between the average scores of the two teams instead of two separate features. From the results of these classifications, it can be stated that the average ranking of the teams is more important than position-specific average rankings, and position-specific average rankings are more important than the maximum rankings in a position. Moreover, the maximum scores for players in each position are more important than the minimum rankings. Additionally, the maximum ranking of forwards in a team is more significant than the maximum scores of midfielders. Based on these observations, the best feature set for classification is presented in Table 19.

The best results from the selected feature subset, shown in Table 20, indicate that 90% of the samples were correctly classified using the SMO and RBF network algorithms. These findings suggest that the feature list presented in Table 19 contains sufficient information to accurately predict match outcomes, thereby confirming the effectiveness of the proposed ranking aggregation method.

Classification is performed using aggregated ranking features to predict match outcomes, combining player scores

TABLE 19. Best subset of ranked features for game outcome classification.

Rank	Feature
1	Average score of the home team
2	Average score of the away team
3	Difference between the average scores of forwards of the home and away teams
4	Difference between the average scores of defenders of the home and away teams
5	Difference between the average scores of goalkeepers of the home and away teams
6	Difference between the average scores of midfielders of the home and away teams
7	Maximum score of forwards in the home team
8	Maximum score of forwards in the away team

TABLE 20. Predicting the outcome with 8 selected characteristics.

Algorithm	Kappa	F-Measure	ROC	Classification accuracy (%)
SMO	0.8451	0.901	0.946	90
RBFNetwork	0.845	0.9	0.97	90
FT	0.8316	0.891	0.966	89.2105
LibSVM	0.807	0.875	0.906	87.6316

from the current and previous matches within a sliding window. A weighted average prioritizes recent performances, ensuring current matches have the most influence. Initial tests with a two-match window gradually extend to ten matches, showing that classification accuracy declines as the window size increases, regardless of whether all 27 features or a reduced set of 8 features are used. The results of match outcome predictions, utilizing player scores from both the current and past matches, are presented in Table 20. The window size represents the number of past matches (including the current one) considered in the aggregated rankings. A weighted averaging method is applied, giving greater significance to recent matches.

TABLE 21. Presents the classification accuracy percentages for predicting match outcomes using aggregated rankings and the SMO algorithm.

Window Size	Using 27 Features	Using 8 Important Features
1	89.7368	90.0000
2	78.4211	79.2105
3	74.7368	74.7368
4	69.2105	71.0526
5	67.8947	68.1579
6	65.5263	65.2632
7	64.7368	64.7368
8	61.8421	62.6316
9	61.8421	61.8421
10	61.0526	61.5789

The differences observed among the various algorithms can be attributed to several key factors. First, the computational complexity of each algorithm plays a significant role in performance, particularly as the size of the dataset increases. Algorithms with higher complexity tend to require more processing time and resources, which can affect efficiency. Second, the sensitivity of the algorithms to variations in input data also contributes to performance discrepancies. Some algorithms may exhibit better robustness to noise and data fluctuations, while others are more susceptible to

such changes. Lastly, the efficiency in terms of convergence rates and memory usage varies across algorithms, which can significantly influence their overall performance in real-world applications. These factors, when considered together, provide a clearer understanding of why certain algorithms perform better under specific conditions.

V. CONCLUSION

The study aimed to identify key performance attributes for players through classification experiments, assigning categories for each position (midfielder, striker, defender, goalkeeper) and using pruning techniques to find optimal feature sets. The most significant attributes yielded mean absolute errors of 0.17 for forwards, 0.28 for defenders, 0.22 for goalkeepers, and 0.18 for midfielders. These attributes were used to generate ''reward and punishment'' rankings, revealing hidden criteria used by experts on WhoScored.com. The second goal was to determine which performance elements impact match outcomes, with a model using 34 features achieving 63.4% accuracy. The third objective was to create an aggregation method for team scores, resulting in 27 team-level features, with 8 aggregated features reaching 90% prediction accuracy. Using a weighted averaging strategy that prioritized recent matches, prediction performance declined as more past matches were included, indicating that current performance has a stronger influence on outcomes. This supports the hypothesis that expert rankings are heavily influenced by recent match results.

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